Contents

Chapter 1

Brownian motion in fluid flows

1.1 Introduction

Fluid flows are ubiquitous in many biological settings. The study of fluid dynamics provides insights on multiple ranges of applications. Oceanic currents and large-scale weather phenomena [3], the lift of an airplane foil [1], surfing [2] and drug biodisponibility all rely at least partly on a range of application of fluid dynamics.

We commonly use the Reynolds number R_e to predict the behaviour of a fluid. This number is a ratio between inertial and viscous forces in the fluid

$$R_\mathrm{e} = \frac{\rho U L}{\eta} , \tag{1.1}$$

where ρ is the fluid's density, U is a characteristic velocity, L is a characteristic length and η is the fluid's dynamic viscosity. Turbulence in a flow occurs at very high Reynolds number $R_\mathrm{e} \gg 1$, whereas a very small Reynolds number $R_\mathrm{e} \ll 1$ is considered laminar, with a smooth and constant profile. A very classic example is that of cigarette smoke (Fig. **??**). The smoke exists the cigarette's tip at very low velocity. Higher up, the smoke rises much faster and the plume is wider. By taking air density at room temperture $\rho_\mathrm{air} = 1.2\mathrm{kg.m}^{-3}$, viscosity at about $\eta_\mathrm{air} = 1.8\mathrm{Pa.s}$, and a smoke characteristic velocity $U_\mathrm{smoke} = 1\mathrm{cm.s}^{-1}$ and characteristic plume width $U_\mathrm{smoke} = 1\mathrm{mm}$, we can estimate the first regime at about $R_\mathrm{e} = 0.5$.

I then want to say that I am interested here in low re flows, becuse they shape the world of microfluidics and have many biological applications. Then, I move to solutions in flows and why they are important to understant. I give the different parts of the chapter.

1.2 Taylor-Aris dispersion

I introduce the classical Taylor-Aris dispersion theory to explain the behaviour of a briwnian solution in a fluid flow. I specify that e work with a perfect gaz solution, and that colloid interactions will be ignored.

1.2.1 Theory

I give here first the equations, the important quantities (peclet number, diffusion, dispersion).

I give all the expected results and profiles and figures

1.2.2 Investigating the diffusion of a Brownian solute in a Poiseuille flow normal to the flow direction

I explain the issues the Paris team were facing. I propose my numerical solution and recall the use of the hydrotools code.

Description of the setup

I describe the paris experimental setup.

Numerical methods

I briefly present my code, but mostly the parametrisation and argumentation for the huge gravity.

Results

I conclude on the paris project and why hydrodynamics are not at pplay there.

Bibliography

[1] J. D. Anderson. *Fundamentals of Aerodynamics*. 7th ed. McGraw-Hill Education, 2016.

[2] T. Butt and P. Russell. *The Physics of Surfing*. 2nd ed. CRC Press, 2024.

[3] J. Pedlosky. *Geophysical Fluid Dynamics*. New York: Springer, 1987.